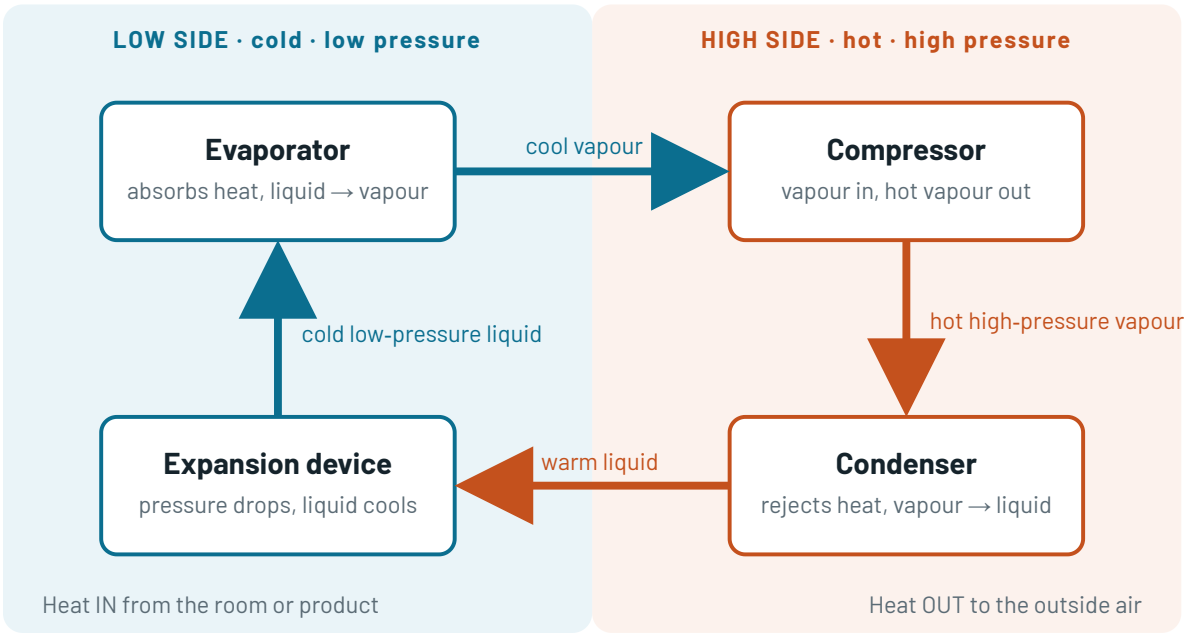
F-Gas Category I in Pictures

38 topics, one per page. Each has the drawing that shows how it works, four boxes with what you need to know, and the one line to remember. Then 30 practice questions, every number on one page, and a plain-English glossary.



Great Britain edition · 2026 · Written for the GB F-Gas Regulation (retained EU 517/2014) and the ODS Regulation · Independent revision material, not official City & Guilds content

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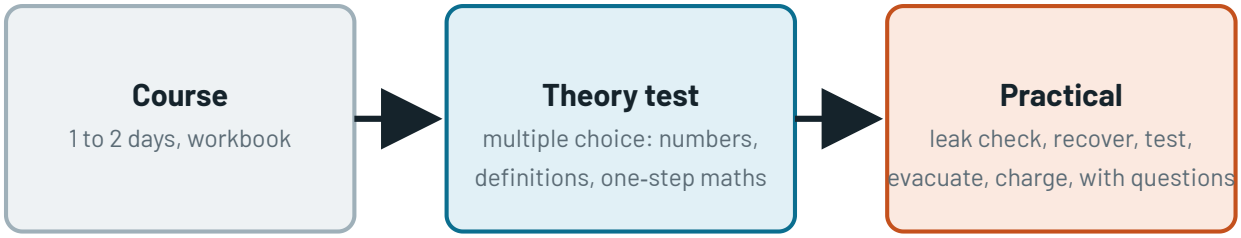
Companion practice app: **fgas-trainer.vercel.app**. The questions and glossary in this book are the same ones the app uses, so they always agree. Tap any dotted word in the app for the glossary entry printed here.

Independent revision material written from the published 2079-11 syllabus and the GB F-Gas and ODS Regulations, using the GWP values the regulation uses (IPCC AR4). Not official City & Guilds content and not affiliated with City & Guilds or any training provider. Where your course workbook or tutor differs, follow them.

START HERE

HOW THE EXAM WORKS

Two tests, one certificate: theory and practical for City & Guilds 2079-11.



Pass both and you hold Category I: every activity, any charge size.

MULTIPLE CHOICE THEORY

Questions on the regulations, refrigerants, the cycle, leak checking, recovery and safety. Mostly numbers and definitions.

PRACTICAL ON A RIG

Leak check, recover, pressure test, evacuate and charge, with the assessor asking questions as you work.

ONE-STEP MATHS

kg × GWP ÷ 1000 gives tonnes CO2e. From that you read off the leak check interval.

CATEGORY I

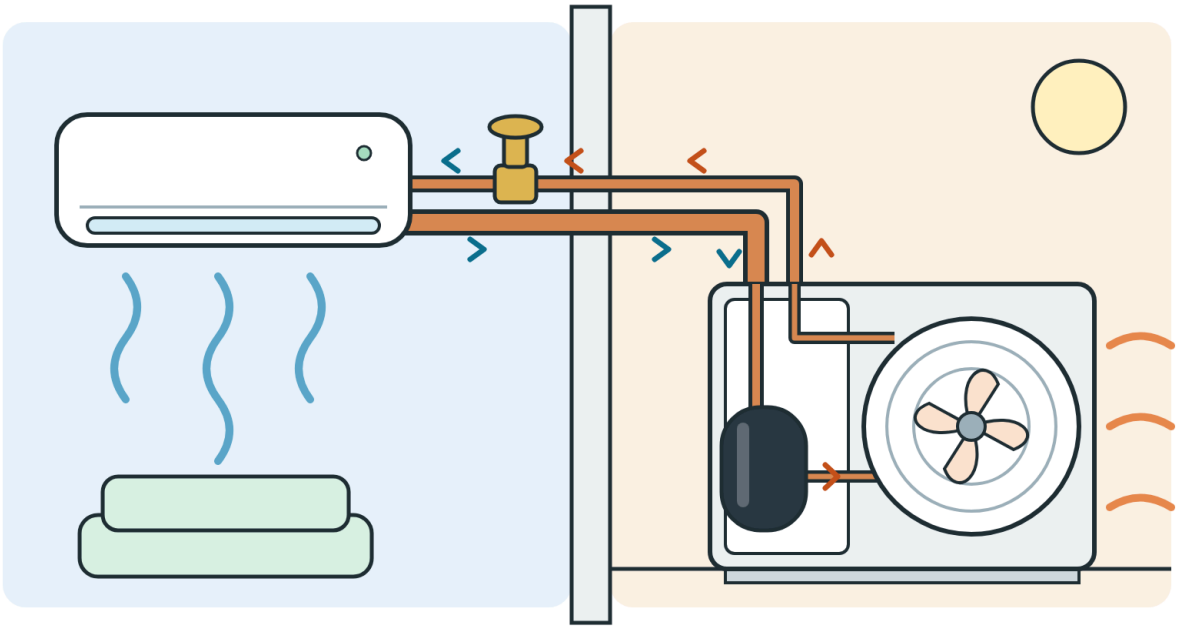
The full ticket: every activity on any charge size. Categories II, III and IV are limited versions.

GREAT BRITAIN RULES

GB kept EU 517/2014 as its own F-Gas Regulation. The newer EU 2024/573 applies in Northern Ireland, not here.

THE CYCLE IN FOUR STAGES

One refrigerant round a closed loop, carrying heat from inside to outside.



1. COMPRESSION

The compressor draws in cool vapour and sends it out hot and at high pressure.

2. CONDENSATION

In the condenser the hot vapour gives its heat to the outside air and becomes warm liquid.

3. EXPANSION

At the expansion device the liquid loses pressure and becomes very cold.

4. EVAPORATION

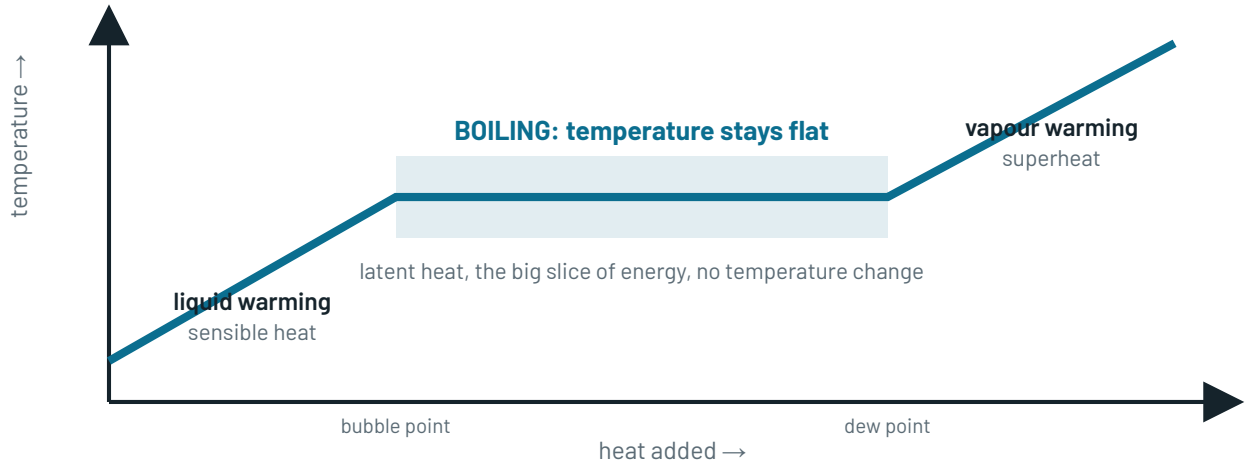
In the evaporator the cold liquid boils, taking heat from the room, and returns as vapour.

TWO SIDES, TWO PRESSURES

Compressor outlet to expansion device is the high side (red gauge). Expansion device to compressor inlet is the low side (blue gauge).

WHY IT GETS COLD

Latent heat: the energy that changes state without changing temperature.



SENSIBLE HEAT

Changes the temperature of something. Warming liquid, warming vapour. You can feel it on a thermometer.

LATENT HEAT

Changes state, liquid to vapour or back, at a constant temperature. Far more energy per kilogram.

PRESSURE SETS THE BOILING POINT

Lower the pressure and the refrigerant boils colder. That is all the expansion device does.

THE PT CHART

For every refrigerant, each pressure matches one saturation temperature. Gauges plus a PT chart tell you the temperature inside.

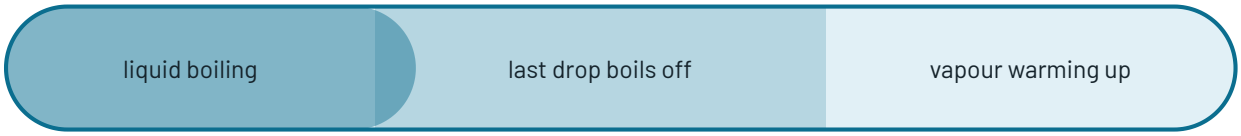
THE WHOLE TRICK

Boil liquid where you want cold (evaporator), condense vapour where you can dump heat (condenser). Latent heat does the carrying.

SUPERHEAT AND SUBCOOLING

Two measurements that tell you what is really happening inside the pipe.

Superheat is measured here



Evaporator → suction line. Saturation temperature from the suction pressure, actual temperature from a probe on the pipe.

Superheat = pipe temperature – saturated evaporating temperature (normal 5 to 10 K)

Subcooling is measured here



Subcooling = saturated condensing temperature – liquid line temperature (normal 3 to 8 K)

SUPERHEAT

Suction line temperature minus the saturation temperature for the suction pressure. Normal 5 to 10 K.

WHY IT MATTERS

Proves every drop of liquid boiled off before the compressor. Zero superheat means liquid is heading for it.

SUBCOOLING

Saturation temperature for the discharge pressure minus the liquid line temperature. Normal 3 to 8 K.

WHY IT MATTERS

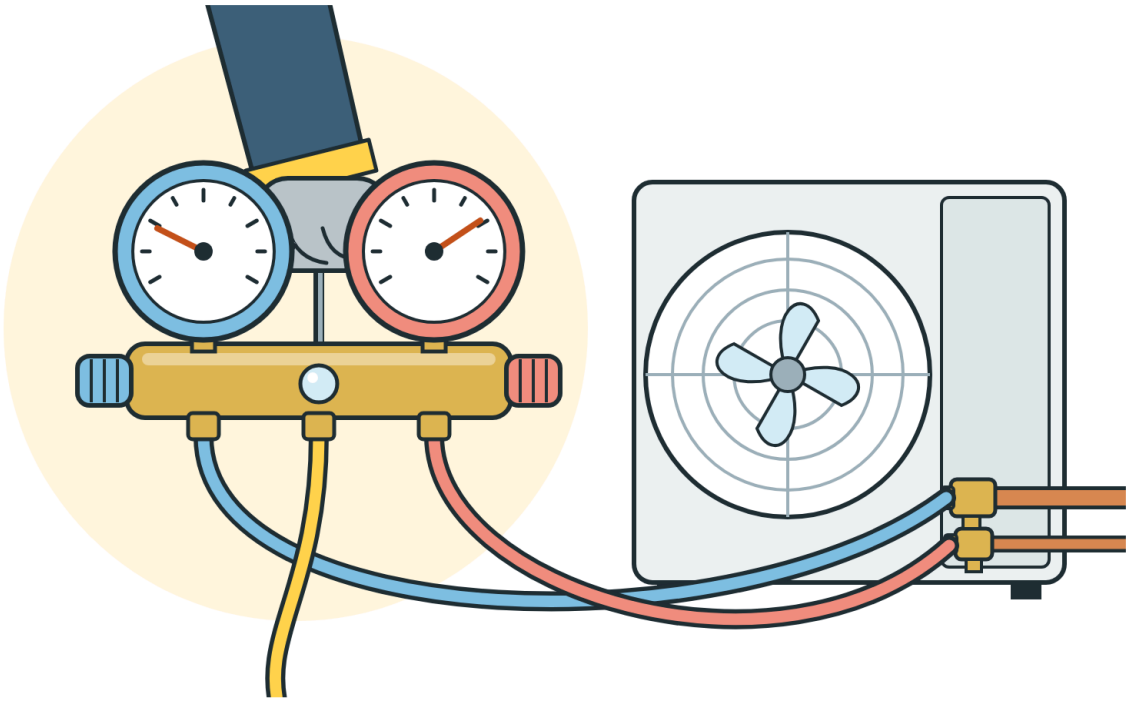
Proves a solid column of liquid is reaching the expansion device. Low subcooling with bubbles means low charge.

ON BLENDS

Use the dew point for superheat and the bubble point for subcooling.

THE MANIFOLD GAUGE SET

The instrument that shows what is happening inside the circuit.



BLUE: LOW SIDE

A compound gauge: reads vacuum and pressure. Connects to the suction service port.

RED: HIGH SIDE

A pressure gauge to around 50 bar. Connects to the discharge or liquid line port.

YELLOW: THE SERVICE HOSE

Centre hose goes to the cylinder, recovery unit or vacuum pump. Purge it before opening a valve.

NOT FOR VACUUM

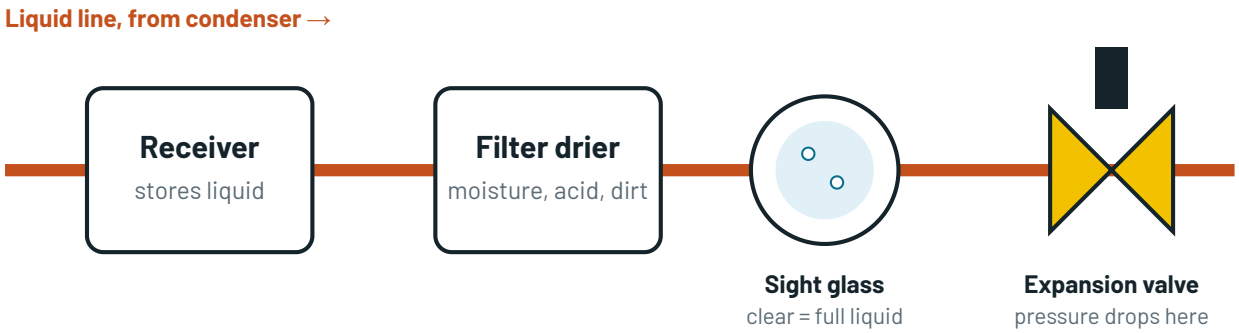
The compound gauge cannot tell 5000 microns from 500. Use an electronic vacuum gauge for evacuation.

EVERY CONNECTION COSTS GAS

Each hose you connect loses a little refrigerant. Use low-loss fittings, keep hoses short and only connect what you need.

THE LIQUID LINE AND ITS PARTS

Between the condenser and the expansion device sit the parts that protect the system.



RECEIVER

Stores liquid after the condenser and takes up changes in demand. Used to hold the charge during pump-down.

FILTER DRIER

Desiccant absorbs moisture and acid, a screen traps debris. Replace it every time the system is opened.

SIGHT GLASS

Clear means full liquid. Bubbles mean flash gas. Many have a moisture indicator that changes colour when wet.

SCHRADER CAPS

The valve core is a service aid. The brass cap and its seal are what stop leaks. Always refit caps.

ALSO ON THE SYSTEM

A suction accumulator catches liquid before the compressor. HP and LP cut-outs stop it on high head pressure or loss of charge.

THE HEAT PUMP AND THE 4-WAY VALVE

The same circuit that cools in summer heats in winter.

WINTER: condenser, heats the room the valve swaps which coil gets the hot gas **WINTER: evaporator, takes heat from outsid**



SUMMER: evaporator, cools the room compressor and expansion device stay the same **SUMMER: condenser, dumps heat outside**

THE REVERSING VALVE

A four-way valve redirects the compressor discharge so either coil can receive the hot gas.

IN WINTER

The indoor coil becomes the condenser and heats the room. The outdoor coil evaporates, pulling heat from cold air.

DEFROSTING

In cold weather the outdoor coil frosts over. The cycle briefly reverses to melt it. The indoor fan pauses: normal, not a fault.

MORE HEAT THAN POWER

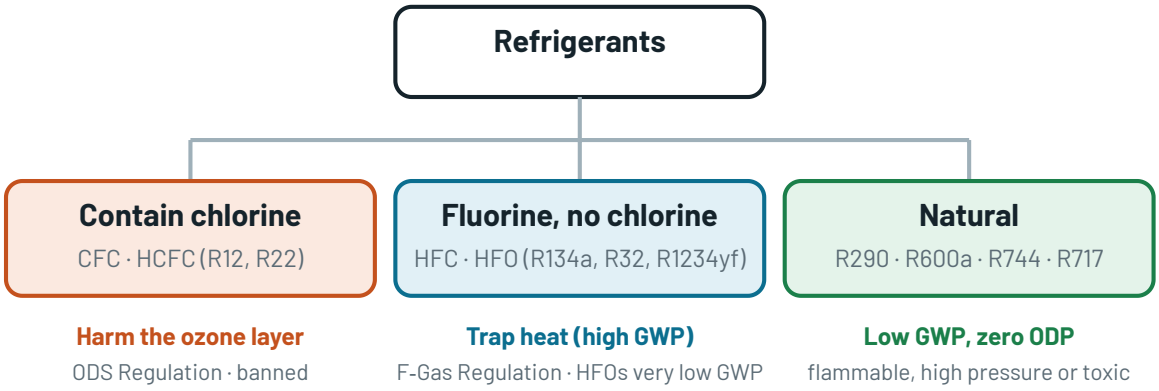
Heat delivered is several times the electricity used, because most of it is moved from outside, not generated.

SAME RULES

A heat pump contains F-gas like any other system: same certification, leak checks, labels and records.

THE REFRIGERANT FAMILIES

Which gases harm the ozone layer, which trap heat, and which law covers each.



Exam trap: R22 is an ozone depleting substance, not an F-gas.

CFC AND HCFC

Contain chlorine, which destroys ozone. R12, R22. Controlled by the ODS Regulation and banned for service.

HFC

Fluorine but no chlorine: zero ODP, high GWP. R134a, R32, R410A, R404A. These are the F-gases.

HFO

Newer synthetics with a very low GWP of about 4. R1234yf, R1234ze. Usually A2L, mildly flammable.

NATURAL

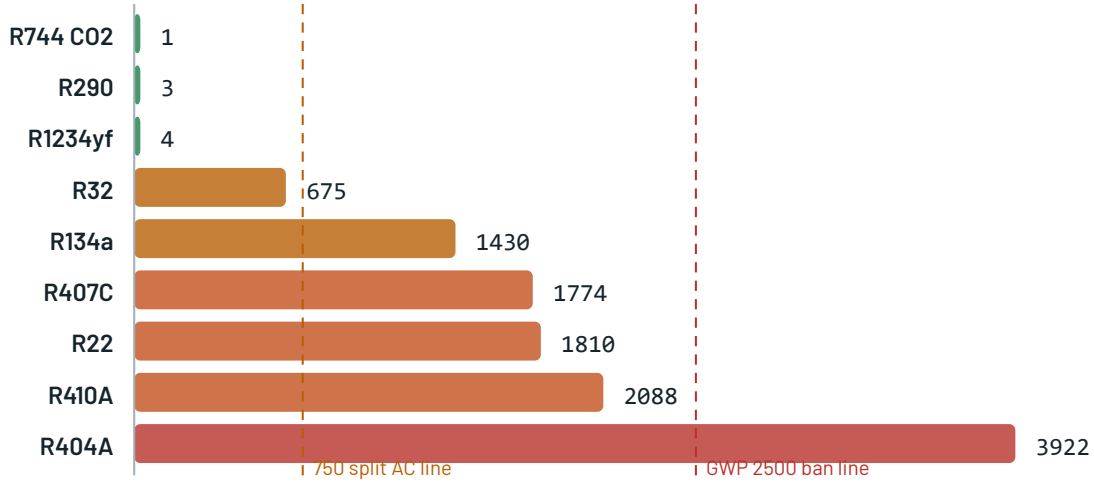
R290 propane, R600a isobutane, R744 CO2, R717 ammonia. Tiny GWP, but flammable, high pressure or toxic.

EXAM TRAP

R22 is an ozone depleting substance under the ODS Regulation. It is not an F-gas.

GWP: THE NUMBERS TO LEARN

Global Warming Potential: how much heat 1 kg traps compared with 1 kg of CO2 over 100 years.



CO2 IS 1

Everything is measured against CO2. Ammonia is 0, propane 3, the HFO R1234yf about 4.

THE BIG FIVE

R32 = 675. R134a = 1430. R407C = 1774. R410A = 2088. R404A = 3922.

THE 2500 LINE

Refrigerants at 2500 or above (R404A) are hit by the 2020 bans on new equipment and virgin servicing.

THE 750 LINE

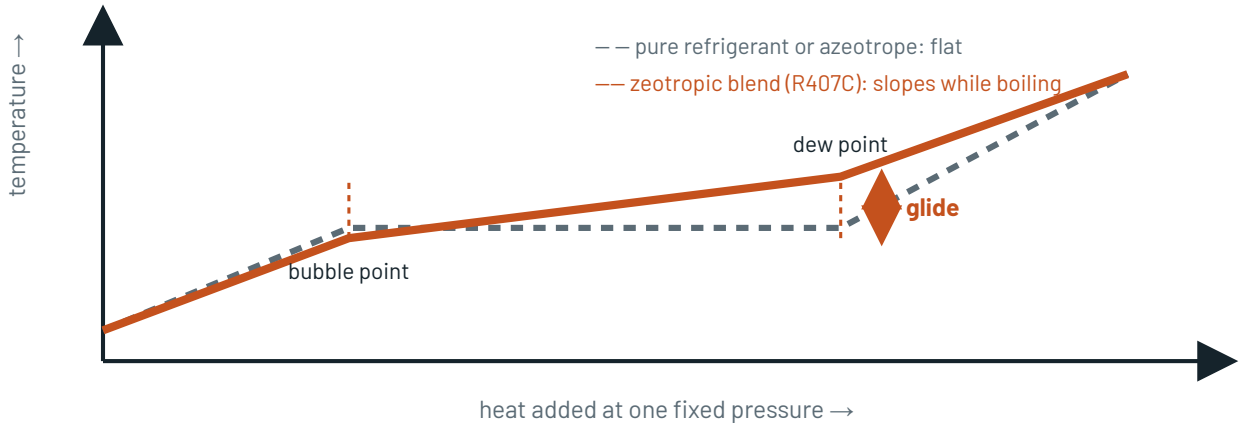
From 2025 single splits under 3 kg may not use refrigerant of GWP 750 or more. R410A is out, R32 is in.

WHICH VALUES

The regulation uses the IPCC Fourth Assessment Report (AR4) figures. Newer reports give slightly different numbers; the exam uses these.

BLENDS AND TEMPERATURE GLIDE

Most modern refrigerants are mixtures, and mixtures do not boil at one temperature.



ZEOTROPIC: R400 SERIES

R407C, R410A, R404A. The parts boil at different temperatures, so the blend has a glide.

BUBBLE AND DEW POINT

Boiling starts at the bubble point and finishes at the dew point. The gap is the glide: about 5 to 7 K for R407C.

CHARGE AS LIQUID

Vapour off the top of the cylinder takes the lighter parts first and leaves the wrong mix behind. Always liquid.

AZEOTROPIC: R500 SERIES

R507A behaves as one substance with no glide. R410A glides so little it is called near-azeotropic.

ON THE GAUGES

Superheat from the dew point, subcooling from the bubble point. Many digital manifolds do this for you once you pick the refrigerant.

THE SAFETY CLASSES

A letter and a number show how toxic and how flammable a refrigerant is.

Letter = toxicity **Number = flammability**

	1 no flame propagation	2L lower flammability, slow flame	2 flammable	3 higher flammability
A lower toxicity	**A1** R134a, R410A, R404A, R744	**A2L** R32, R1234yf, R454B	**A2** R152a	**A3** R290, R600a
B higher toxicity	**B1** R123	**B2L** R717 ammonia	**B2** rare	**B3** rare

A1

Lower toxicity, no flame propagation. R134a, R410A, R404A, R744. Still an asphyxiation and frostbite hazard.

A2L

Lower toxicity, mildly flammable with a slow flame. R32, R1234yf, R454B. Spark-free tools and rated equipment.

A3

Lower toxicity, highly flammable. R290 propane, R600a isobutane. Treat like LPG, strict charge limits.

THE LETTER B

Higher toxicity. Ammonia R717 is B2L: toxic and mildly flammable, but its smell gives early warning.

MORE FLAMMABLE, MORE RULES

Maximum charge, room size, ventilation and tools depend on the class. Follow EN 378 and the manufacturer.

OIL AND MOISTURE

HFC systems use POE oil, and POE oil loves water. That is where most contamination starts.

Moisture gets in → open pipe, wet POE oil, poor vacuum

Acid forms → water + refrigerant + POE + heat

Copper plating → copper dissolves, plates bearings

Compressor fails → burnt windings, seized

Break the chain early: cap pipes, seal oil, new drier, evacuate below 500 microns.

POE OIL

Polyol ester oil for HFCs. Mineral oil does not mix with them. Keep containers sealed.

HYGROSCOPIC

POE pulls moisture out of the air in minutes. Cap open pipes, keep systems open as briefly as possible.

ACID

Water plus refrigerant plus POE plus heat makes acid, which attacks motor windings.

COPPER PLATING

Acid dissolves copper, which plates onto hot steel bearing surfaces and seizes the compressor.

BREAK THE CHAIN

New drier every time the system is opened, evacuate below 500 microns, watch the moisture indicator.

RECOVERED, RECYCLED, RECLAIMED

Three words the exam loves, and they mean three different things.

RECOVERED
taken out of the system

in a cylinder, nothing else done

RECYCLED
filtered, oil and water removed

reuse only in the same owner's kit

RECLAIMED
reprocessed to virgin spec

can be sold and used anywhere

MIXED refrigerants cannot be reclaimed: destruction, at your cost

RECOVERED

Removed from the system into a cylinder. Nothing has been done to it yet.

RECYCLED

Basic cleaning: oil separation, filtering, moisture removal. May only go back into the same owner's equipment.

RECLAIMED

Reprocessed at a specialist facility to a virgin standard (AHRI 700). Can be sold and used anywhere.

MIXED

Two refrigerants in one cylinder cannot be separated economically. Destruction only, and you pay.

IDENTIFY FIRST

Check the label and the pressures before you recover. One wrong cylinder ruins a whole batch.

TWO REGULATIONS

One protects the climate, the other protects the ozone layer.

F-GAS

GB version of EU 517/2014
HFCs: R134a, R32, R410A, R404A
cuts greenhouse gas emissions

ODS

GB version of EC 1005/2009
CFCs and HCFCs: R12, R22
protects the ozone layer

Enforced in GB by the Environment Agency, Natural Resources Wales and SEPA

F-GAS REGULATION

GB version of EU 517/2014. Certification, leak checks, records, labels, phase-down and bans for HFCs.

ODS REGULATION

GB version of EC 1005/2009. Banned virgin R22 for service from 2010 and recycled R22 from 1 January 2015.

WHO ENFORCES

Environment Agency in England, Natural Resources Wales, SEPA in Scotland. The HSE covers safety and pressure systems.

NO VENTING

Deliberate release is an offence. Take every feasible precaution against accidental release.

R22 TODAY

A system may run on its existing R22 charge but cannot be topped up with anything. Leak means replace or convert.

PERSONNEL CATEGORIES

Four personnel certificates. Category I is the only one without limits.

	Leak check	Recovery	Install / service
Cat I	Any	Any	Any
Cat II	Any	< 3 kg*	< 3 kg*
Cat III	No	< 3 kg*	No
Cat IV	Any	No	No

* or under 6 kg if hermetically sealed and labelled. Leak check in Cat II and IV means without breaking into the circuit.

CATEGORY I

Leak checking, recovery, installation, maintenance, servicing, repair and decommissioning on any charge size.

CATEGORY II

Leak checks (no break-in) on any system. Everything else only under 3 kg, or 6 kg hermetically sealed and labelled.

CATEGORY III

Recovery only, from systems under 3 kg (6 kg hermetic). No installation or maintenance.

CATEGORY IV

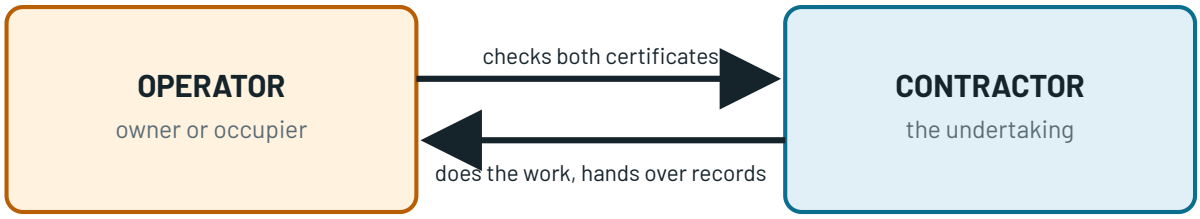
Leak checking only, and only by methods that do not break into the circuit.

HERMETICALLY SEALED

All joints brazed, welded or similar, service ports allowed, tested leak rate under 3 g per year. Only counts if labelled as such.

OPERATOR AND CONTRACTOR

The engineer does the work. The operator carries the legal duty.



legally responsible for leak checks, records, repairs, certified people

company certificate (REFCOM etc.) + engineer's personnel certificate

THE OPERATOR

Whoever has real control over how the equipment runs. Usually the owner or the business on the premises.

OPERATOR DUTIES

Prevent leaks, repair them promptly, have checks done on time by certified people, keep records five years.

COMPANY CERTIFICATE

The business itself must be certified by REFCOM, Quidos or Bureau Veritas, as well as its engineers.

CHECK BEFORE WORK

An operator must see the company certificate and the engineer's personnel card before letting them start.

PRE-CHARGED EQUIPMENT

A split system may only be sold to an end user with evidence that a certified undertaking will install it.

THE EQUIPMENT LABEL

Every system containing F-gas must say so, and say how much.

CONTAINS FLUORINATED GREENHOUSE GASES

Refrigerant	**R410A**
Charge	**7.5 kg**
CO2 equivalent	**15.66 tonnes**
GWP	**2088** hermetically sealed ☐

WHAT IT SAYS

Contains fluorinated greenhouse gases, the refrigerant designation, the GWP.

HOW MUCH

The charge in kilograms and in tonnes CO2 equivalent.

UPDATE IT

Add gas on site and the label must be updated with the new total. An extra label is fine.

HERMETIC

If the equipment is hermetically sealed the label must state it, or the relaxed thresholds do not apply.

WHY THE LABEL MATTERS

The leak check interval and the record duty both depend on the figure printed here.

PHASE-DOWN AND BANS

HFCs are being squeezed out by quota, and specific uses are banned by date.



THE QUOTA

Producers and importers get a shrinking yearly allowance in tonnes CO2e. High-GWP gas becomes scarce and expensive.

2015

Recycled R22 banned. No HFC of GWP 150 or more in new domestic fridges and freezers.

2020

No new stationary refrigeration with HFC of GWP 2500 or more. No virgin GWP 2500+ for servicing systems of 40 t CO2e or more.

2025 AND 2030

2025: single splits under 3 kg must be below GWP 750. 2030: recycled and reclaimed GWP 2500+ also banned for service.

BELOW MINUS 50

Equipment designed to cool products below minus 50 degrees C is exempt from the 2500 GWP bans.

CO2 EQUIVALENT

Every threshold is in tonnes of CO2 equivalent, not kilograms.

kg of refrigerant × GWP ÷ 1000 = tonnes CO2e

Example: 12 kg of R410A → 12 × 2088 = 25,056 kg → 25.1 tonnes CO2e → leak check every 12 months

THE FORMULA

kg of refrigerant × GWP ÷ 1000 = tonnes CO2e.

WORKED EXAMPLE

12 kg of R410A: 12 × 2088 = 25,056 kg = 25.1 t. That is in the 5 to 50 t band: every 12 months.

THE OTHER WAY

5 t is reached by 1.27 kg of R404A, 2.39 kg of R410A, 3.50 kg of R134a or 7.41 kg of R32.

WHY

A small charge of high-GWP gas does more damage than a big charge of low-GWP gas, so it is checked sooner.

ONE LINE OF MATHS

Learn the GWPs and the exam maths is a multiplication and a division.

HOW OFTEN TO CHECK

Three bands, doubled if a fixed leak detection system is fitted.

Charge in tonnes CO2e → how often a certified person must leak check

under 5 t	**No set interval. Still prevent and repair leaks.**
5 t or more	**Every 12 months** 24 months with a leak detection system
50 t or more	**Every 6 months** 12 months with a leak detection system
500 t or more	**Every 3 months** 6 months with detection. Detection system compulsory.

Hermetically sealed and labelled: threshold is 10 t instead of 5 t. Leak detection systems must themselves be checked every 12 mont

5 TONNES OR MORE

Every 12 months. 24 months with a leak detection system.

50 TONNES OR MORE

Every 6 months. 12 months with a leak detection system.

500 TONNES OR MORE

Every 3 months, 6 with detection. A fixed leak detection system is compulsory.

HERMETIC AND LABELLED

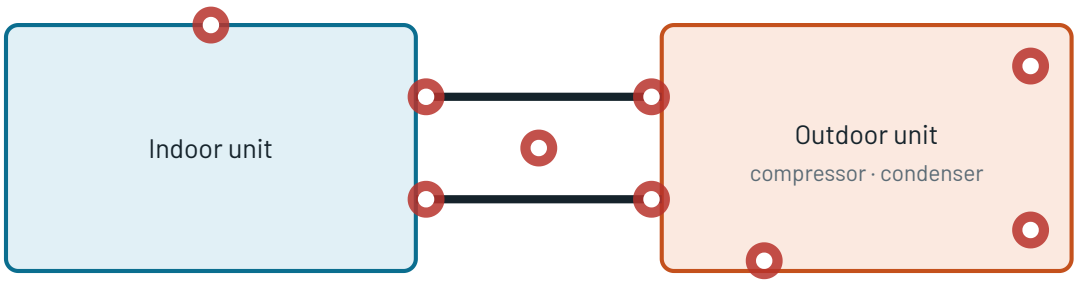
Threshold starts at 10 tonnes instead of 5.

AFTER A REPAIR

Re-check within one month. Leak detection systems themselves are checked every 12 months.

WHERE LEAKS HAPPEN

Refrigerant carries oil with it, so a leak usually leaves a stain.



Red dots: where leaks usually are. Look for oil staining first.

Flare nuts, brazed joints, Schrader cores and missing caps, valve stems, the compressor shaft seal, pipes rubbing on brackets.

FLARE NUTS

The most common leak on splits. Under-tightened, over-tightened or cracked flares.

BRAZED JOINTS

Pinholes and cold joints, often where a joint was rushed or not purged.

SCHRADER CORES

Weeping cores and missing caps. The cap is the seal.

SEALS AND STEMS

Compressor shaft seals, service valve stems and pipes rubbing on brackets.

LOOK FOR OIL FIRST

An oily, dusty patch tells you where to put the probe before you switch it on.

DIRECT AND INDIRECT METHODS

Indirect readings point to a leak. Direct methods find it.

DIRECT: test the pipework

- Electronic detector (≥ 5 g/yr, checked yearly)
- Bubble solution on joints
- UV dye and lamp
- OFN standing pressure test

INDIRECT: read the system

- Pressures and temperatures
- Superheat and subcooling
- Compressor running current
- Oil level, liquid level, sight glass

Indirect readings point to a leak; a direct method finds it; bubbles confirm it.

ELECTRONIC DETECTOR

Sensitivity at least 5 g per year, checked every 12 months. Move slowly under joints, because the gas sinks.

BUBBLE SOLUTION

Cheap, reliable and the way to confirm a hit from the electronic detector.

UV DYE

Dye in the oil glows under a UV lamp wherever it has escaped. Good for slow leaks.

INDIRECT

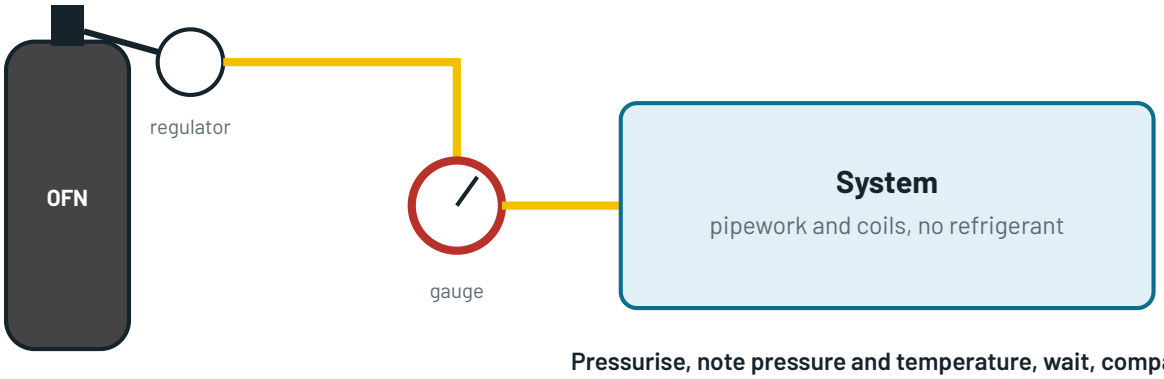
Pressures, temperatures, superheat, running current, oil and liquid levels. A change says something is wrong.

THEN REPAIR IT

Topping up a known leaker instead of fixing it is an offence. Sealants are not a repair.

THE NITROGEN STANDING PRESSURE TEST

Before the vacuum, prove the circuit holds pressure.



Pressurise, note pressure and temperature, wait, compare.

DRY NITROGEN ONLY

Never oxygen or compressed air. Oxygen explodes with oil, air brings moisture. Always through a regulator.

THE PRESSURE MUST NOT DROP

Hold at the manufacturer's test pressure and watch over time.

ALLOW FOR TEMPERATURE

Gas pressure falls as it cools overnight. Compare corrected readings before calling it a leak.

JOINTS WITH FOAM

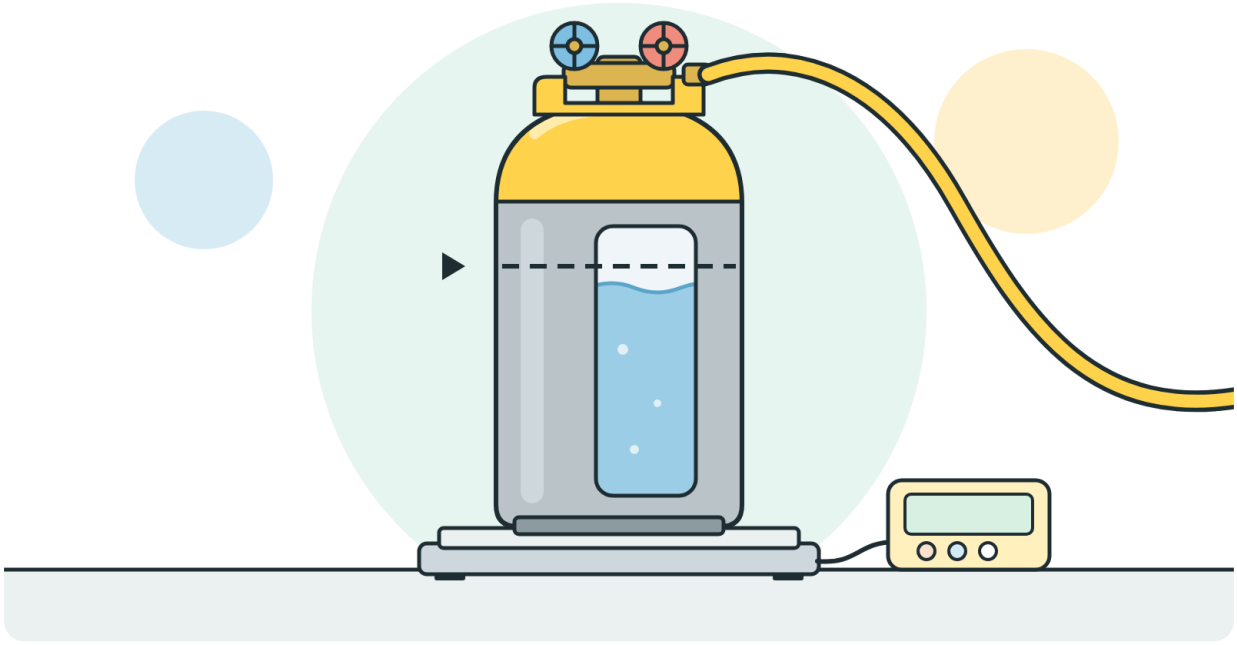
Bubble solution on every joint shows even tiny leaks while the system is under pressure.

AT THE END

Release the nitrogen to open air (it is harmless), then pull the vacuum.

THE RECOVERY CYLINDER

A pressure vessel with rules. Weigh it, label it, never overfill it.



80% BY WEIGHT

On scales, throughout. Liquid expands when warm; an overfilled cylinder can burst.

ONE REFRIGERANT

Dedicated and labelled. Mixed gas cannot be reclaimed.

IN TEST DATE

Cylinders are periodically pressure tested. Check the stamp before you connect it.

EVACUATED FIRST

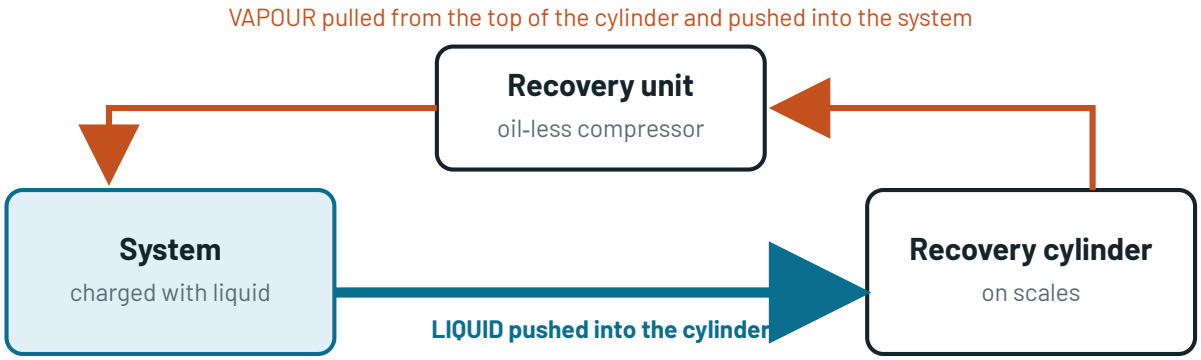
Pull a vacuum on an empty cylinder so it holds no air to contaminate the gas.

NEVER A DISPOSABLE

Disposable cylinders are not built or tested for refilling. Refilling one is dangerous and illegal.

RECOVERY METHODS

Vapour is slow, liquid is fast, push-pull is fastest for big charges.



Fast for large charges. Finish with normal vapour recovery to pull the system below atmospheric pressure.

VAPOUR RECOVERY

From the suction port into the recovery unit. Simple, slow, and how every recovery finishes.

LIQUID RECOVERY

From the liquid line or receiver. Moves far more refrigerant per minute.

PUSH-PULL

Vapour pulled off the cylinder and pushed into the system drives liquid out into the cylinder.

FLAMMABLE GAS

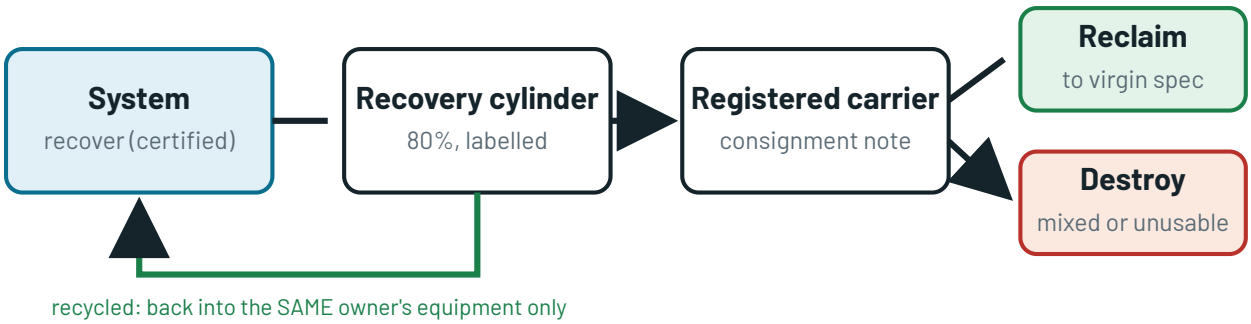
A2L and A3 need a recovery unit rated for flammables, ventilation and no ignition sources.

HOW FAR DOWN

Finish below atmospheric pressure, a slight vacuum, so almost nothing escapes when you open the circuit. Break the vacuum with OFN before brazing.

WHERE THE GAS GOES

Recovered refrigerant and old oil are hazardous waste with a paper trail.



Refrigerant and used oil are hazardous waste. Keep consignment notes for 3 years.

REGISTERED CARRIER

Only a registered waste carrier may take it, to a permitted facility.

CONSIGNMENT NOTE

A hazardous waste consignment note travels with it. Both parties keep copies for three years.

RECLAIM OR DESTROY

Clean single refrigerants are reclaimed. Mixed or contaminated gas is destroyed.

THE OIL TOO

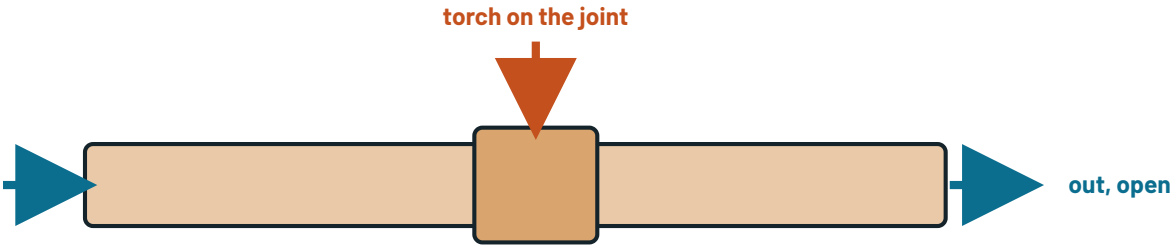
Used compressor oil holds dissolved refrigerant and acid. Sealed container, hazardous waste route.

RECORD IT

The quantity recovered and what happened to it goes in the equipment record, including at decommissioning.

BRAZING WITH NITROGEN

A clean joint on the outside can still be ruining the system on the inside.



OFN in, gentle flow

through a regulator, barely felt on the back of the hand

Nitrogen displaces the air, so no black copper-oxide scale forms inside the pipe to block driers and expansion valves.

NITROGEN INSIDE THE PIPE

A gentle OFN flow displaces the air so no black copper-oxide scale forms in the bore.

WHY IT MATTERS

Scale breaks off and blocks driers, strainers and expansion valves, sometimes months later.

THE RIGHT ALLOY

Copper-phosphorus rods are self-fluxing on copper to copper. Copper to brass or steel needs silver solder and flux.

PROTECT THE SURROUNDINGS

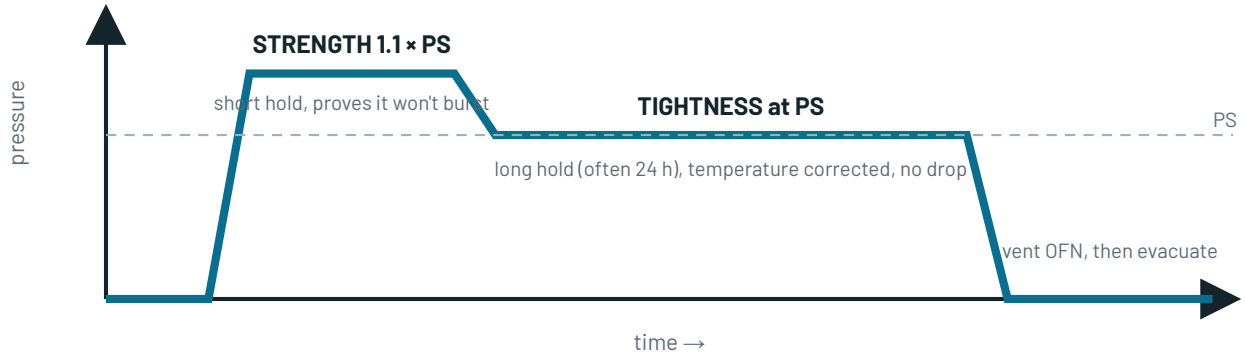
Heat-resistant mat, extinguisher to hand, hot work permit and a fire watch afterwards.

NEVER ON A CHARGED CIRCUIT

Recover first, purge with OFN, prove zero pressure. Heat plus HFC makes hydrogen fluoride.

STRENGTH AND TIGHTNESS TESTS

Prove it will not burst, then prove it does not leak, before any refrigerant goes in.



STRENGTH TEST

OFN at 1.1 times the maximum allowable pressure (PS). A short hold proving the pipework will not fail.

TIGHTNESS TEST

OFN at PS, held long enough (often 24 hours) and temperature corrected, with no pressure drop.

REGULATOR ALWAYS

A nitrogen cylinder is around 200 bar. The regulator is what stands between that and your system.

NEVER

Oxygen, compressed air or refrigerant on its own. Oxygen plus oil explodes.

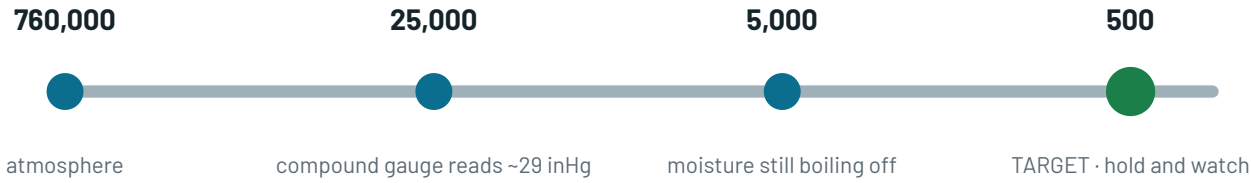
THEN VENT

Release the nitrogen outdoors and move on to evacuation.

THE VACUUM

It removes the air and moisture the refrigerant cannot tolerate.

Vacuum in microns (absolute pressure). Lower is deeper.



Vacuum rises fast after isolating the pump = leak. Creeps up and stalls = moisture still present. Stays put = dry and tight.

WATER BOILS COLD

Under a deep vacuum water boils at room temperature and the pump draws it away.

MEASURED WITH A MICRON GAUGE

The manifold's compound gauge is not enough. An electronic vacuum gauge at the system, below 500 microns.

THE RISE TEST

Isolate the pump and watch. Rising fast is a leak. Creeping and stalling is moisture. Steady is good.

TRIPLE EVACUATION

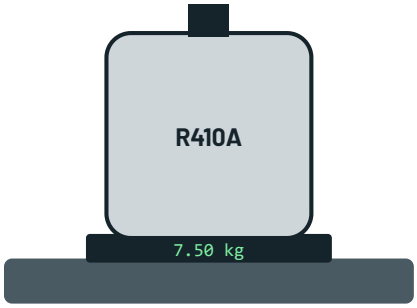
Wet system? Pull, break with OFN, pull, break, pull. The nitrogen soaks up the last moisture.

NEVER START THE COMPRESSOR UNDER VACUUM

The motor windings can arc and fail with no gas to insulate and cool them. Charge first.

CHARGING BY WEIGHT

The nameplate tells you the charge. The scales tell you when you have reached it.



Charge by weight, not by feel

nameplate charge + manufacturer's allowance per extra metre of pipe

zeotropic blends: LIQUID from the cylinder (dip tube or inverted)

liquid into the liquid line with the system off, or metered into suction

then prove it with superheat and subcooling

never slug liquid into a running compressor

BY WEIGHT

Nameplate charge plus the manufacturer's allowance per metre of extra pipe. Scales, not guesswork.

BLENDS AS LIQUID

Zeotropic blends come out of the cylinder as liquid, via the dip tube or by inverting it.

WHERE IT GOES IN

Liquid into the liquid line with the system off, or metered slowly into the suction with it running.

THEN PROVE IT

Run the system and confirm with superheat and subcooling. They confirm the charge; they do not set it.

NEVER SLUG THE COMPRESSOR

Liquid cannot be compressed. A slug of it breaks valves and washes the oil off the bearings.

COMMISSIONING IN ORDER

Six steps, always in this sequence.



OFN for the first two, vacuum pump and micron gauge for the third, scales for the fourth, thermometer and gauges for the fifth.

1 AND 2: TEST

Strength test at 1.1 × PS, then tightness test at PS, both with OFN.

3: EVACUATE

Below 500 microns on a micron gauge, hold, and pass the rise test.

4: CHARGE

By weight, blends as liquid, cores and caps refitted.

5 AND 6: RUN, LABEL, RECORD

Check superheat and subcooling, update the label, write up the record.

WHY THIS ORDER

Leaks are found before gas goes in. Air and water are removed before gas goes in. Everything after that is proof.

FOUR FAULT FINGERPRINTS

Read suction, head, superheat and subcooling together, not one at a time.

Low charge	Overcharge	Air in system	Restriction
Suction ↓	Suction ↑	Suction –	Suction ↓
Head ↓	Head ↑	Head ↑↑	Head ↓
Superheat ↑	Superheat ↓	Superheat –	Superheat ↑
Subcool ↓	Subcool ↑	Subcool –	Subcool –
bubbles in glass	high amps, floodback	head above PT chart	temp drop across drier

LOW CHARGE

Low suction, low head, high superheat, low subcooling, bubbles in the glass. Find the leak before adding gas.

OVERCHARGE

High head, high subcooling, low superheat, high current. Liquid may flood back to the compressor.

AIR IN THE SYSTEM

Head pressure higher than the PT chart predicts for the liquid temperature. Poor evacuation.

RESTRICTION

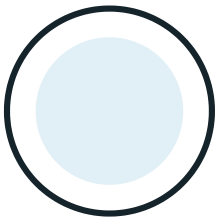
Low suction, high superheat, normal subcooling, a temperature drop across the blocked drier or valve.

LOW CHARGE OR RESTRICTION?

They look alike on the gauges. Subcooling splits them: low for low charge, normal for a restriction.

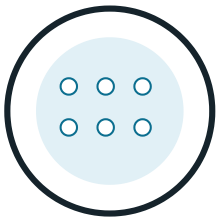
READING THE SIGHT GLASS

A small window that answers two questions: is it full, and is it dry?



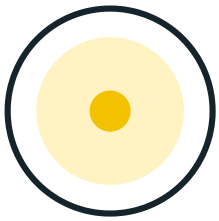
CLEAR

full column of liquid: good



BUBBLES

flash gas: low charge or restriction



WET

indicator changed colour: change the drier

CLEAR

A full column of liquid reaching the expansion device. Confirm with subcooling.

BUBBLES

Flash gas in the liquid line: low charge, or a pressure drop across a blocked drier.

CHECK THE DRIER

A drier colder on its outlet than its inlet is blocked and acting as an expansion device.

WET INDICATOR

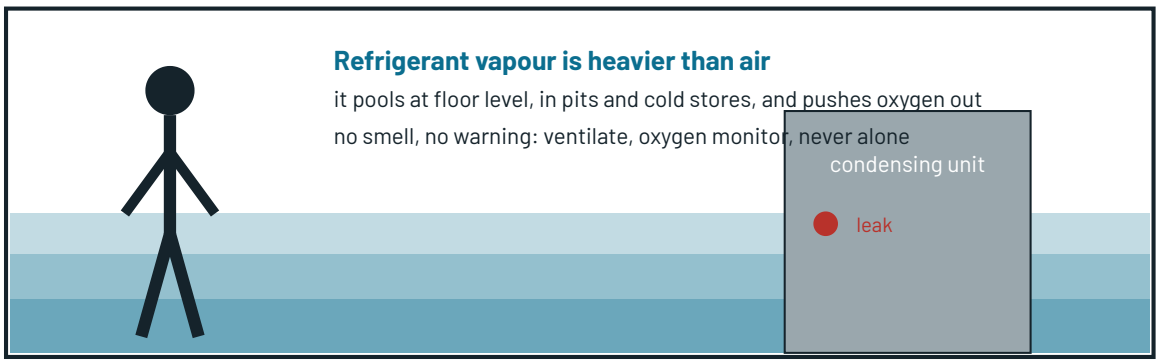
Colour change means moisture. New drier, proper evacuation, find the entry point.

THE WET SYSTEM

Intermittent cooling as ice forms and clears at the expansion device, acidic oil, and eventually a dead compressor.

HEAVIER THAN AIR

Refrigerant has no smell and sinks. The danger is the oxygen it pushes out.



Plant rooms over a set charge need fixed detection and alarms under EN 378.

IT POOLS LOW

Floors, pits, cold stores and plant rooms. Ventilate before entering.

NO WARNING

Odourless and colourless. Use an oxygen monitor or fixed detection. Never alone in a suspected leak.

FROSTBITE

Liquid flashes off well below zero and freezes skin and eyes. Gloves and goggles on every connection.

FIRST AID

Fresh air, warm (not hot) water on cold burns, medical help. The safety data sheet tells you more.

COSHH

Every refrigerant has a safety data sheet. Your employer must assess and control the risk from it.

CYLINDERS: DO AND NEVER

A refrigerant cylinder is a pressure vessel. Treat it as one.

DO

- ✓ Keep upright and secured
- ✓ Fit valve caps for transport
- ✓ Store cool, ventilated, out of sun
- ✓ Warm with heater belt or water < 40 °C
- ✓ Check the test date

NEVER

- ✗ Heat with a naked flame
- ✗ Fill a recovery cylinder past 80%
- ✗ Refill a disposable cylinder
- ✗ Mix refrigerants in one cylinder
- ✗ Use oxygen or air to pressurise

UPRIGHT AND SECURED

Strapped so it cannot fall, valve cap fitted for transport, under ADR rules in the van.

KEEP IT COOL

Out of direct sun, below 50 degrees C, in a ventilated space.

WARMING TO CHARGE

Heater belt or warm water under 40 degrees C only. Never a flame.

NEVER

Flame on a cylinder, past 80% in a recovery cylinder, refilling a disposable, mixing gases, oxygen or air for testing.

PSSR

Refrigeration systems with compressor motor power above 25 kW need a written scheme of examination under the Pressure Systems Safety Regulations.

FLAMES AND REFRIGERANT

Heat turns a harmless HFC into a poison. Hot work has rules for a reason.



Pipe still holding refrigerant

HEAT + HFC = HYDROGEN FLUORIDE

toxic, corrosive gas; plus molten metal blown out of the joint

BEFORE ANY FLAME

recover · purge with OFN · prove zero pressure
clear flammables · extinguisher · hot work permit
fire watch after you finish

HYDROGEN FLUORIDE

HFCs break down in a flame or on a hot surface into toxic, corrosive gas.

MAKE SAFE FIRST

Recover, purge with OFN, prove zero pressure before any torch touches the pipe.

HOT WORK PERMIT

Site authorisation listing the precautions: flammables cleared, extinguisher, fire watch.

NO SMOKING

A cigarette in a plant room with a leak makes the same gas. Keep ignition sources out.

PRESSURE PLUS HEAT

Brazing a line still under pressure blows molten metal out of the joint.

WORKING WITH A2L REFRIGERANTS

R32 is in most new splits. Mildly flammable means some new habits.



R32 · R1234yf · R454B

- Needs a rich leak AND a spark to burn
- Slow flame, but it does burn
- No ignition sources in the work area
- Spark-free tools, rated recovery unit and detector
- Charge limits by room size: EN 378
- Ventilate before and during work

WHAT A2L MEANS

Lower toxicity, lower flammability, slow flame. It needs a rich leak and a spark, but it will burn.

NO IGNITION SOURCES

No naked flames, no sparking tools or switches in the work area, no smoking.

RATED EQUIPMENT

Recovery unit, vacuum pump and leak detector rated for A2L. Spark-free tools.

CHARGE LIMITS

EN 378 limits the charge by room size so a leak cannot reach a flammable concentration.

VENTILATE

Before and during work, and check for leaks before any hot work.

THE EQUIPMENT RECORD

Five years, both parties, and exactly these items.

F-GAS EQUIPMENT RECORD — keep 5 years · operator and contractor

Equipment	Cold room pack, plant room 2
Refrigerant / charge	R404A · 30 kg · 117.7 t CO2e
Leak check interval	6 months (no fixed detection)
Date · result · engineer	08/11/2026 · no leak · J. Smith, Cat I, Acme Ltd (REFCOM)
Gas added	0 kg (last top-up 2.0 kg reclaimed, 03/2026, from X Reclaim Ltd)
Gas recovered	—
Decommissioning	—

WHAT IS IN IT

Refrigerant type and charge, gas added and why, gas recovered, who did the work, leak check dates and results.

RECYCLED OR RECLAIMED

If reused gas went in, the name and address of the recycling or reclamation facility.

FIVE YEARS

Kept by the operator and by the contractor, available to the regulator on request.

AT THE END

Decommissioning: the measures taken to recover and dispose of the refrigerant.

WHICH EQUIPMENT

Anything subject to leak checks: 5 tonnes CO2e or more, 10 for labelled hermetic. Consignment notes are kept 3 years, not 5.

XI Practice questions

Thirty exam-style questions. Mark yourself with the key that follows, then read the explanation for anything you missed.

1.

REGULATIONS & CERTIFICATION

What is the main purpose of the F-Gas Regulation?

- A To reduce emissions of fluorinated greenhouse gases and limit their use
- B To protect the ozone layer from chlorine-based refrigerants
- C To set minimum energy efficiency ratings for air conditioning
- D To control the price of refrigerant cylinders

2.

REGULATIONS & CERTIFICATION

A Category IV certificate permits which activity?

- A Brazing of refrigerant pipework
- B Leak checking only, without breaking into the refrigerant circuit
- C Recovery from systems under 3 kg
- D Installation of small split systems

3.

REGULATIONS & CERTIFICATION

How does the HFC phase-down work?

- A All HFC refrigerants are banned from a single date
- B Only refrigerants with a GWP above 2500 are restricted
- C A quota system limits the amount of HFC, measured in CO2 equivalent, that producers and importers may place on the market each year
- D Each contractor is allowed a fixed number of kilograms per year

4.

REGULATIONS & CERTIFICATION

What must the label on equipment containing F-gas state?

- A Only the refrigerant name
- B The installer's phone number and the date of installation
- C The energy rating of the equipment
- D That it contains F-gas, the refrigerant designation, the charge in kg and its CO2 equivalent, and the GWP

5.

REGULATIONS & CERTIFICATION

Which of these best describes a hermetically sealed system under the F-Gas Regulation?

- A All refrigerant-containing parts are made tight by welding, brazing or a similar permanent connection, and the tested leak rate is below 3 g per year
- B Any system with a sealed compressor
- C A system that has never been opened since manufacture
- D A system with no service ports

6.

REFRIGERANTS & GWP

What is the Global Warming Potential (GWP) of carbon dioxide (R744)?

- A 10
- B 1
- C 0
- D 100

7.

REFRIGERANTS & GWP

What is the GWP of R32?

- A 2088
- B 4
- C 675
- D 1430

8.

REFRIGERANTS & GWP

Which of these is an HFO refrigerant?

- A R410A
- B R290
- C R717
- D R1234yf

9.

REFRIGERANTS & GWP

What is 'temperature glide'?

- A The difference between the bubble point and dew point of a zeotropic blend at a given pressure
- B The drop in temperature across the expansion valve
- C The rise in temperature across the compressor
- D The change in cylinder temperature while charging

10.

REFRIGERANTS & GWP

Why must different refrigerants never be mixed in a system or a recovery cylinder?

- A Because it voids the cylinder warranty
- B Mixed refrigerant cannot be reclaimed, must be sent for destruction, and gives unpredictable pressures and performance
- C It is permitted as long as the GWPs are similar
- D Because the blend will become flammable

11.

LEAK CHECKING

Equipment containing 5 tonnes CO2e or more of F-gas (but less than 50) must be leak checked at least every:

A 3 months

B 24 months in all cases

C 12 months (24 months if a fixed leak detection system is fitted)

D 6 months

12.

LEAK CHECKING

After a leak has been repaired, the equipment must be checked again to confirm the repair worked within:

A 1 week

B 6 months

C 12 months

D 1 month

13.

LEAK CHECKING

A system has 12 kg of R410A and no fixed leak detection. How often must it be leak checked?

A Every 12 months (12 x 2088 = 25 tonnes CO2e)

B Every 6 months

C Every 3 months

D No check required

14.

LEAK CHECKING

A nitrogen standing pressure test shows a small drop in pressure overnight. What must you consider before declaring a leak?

A Nothing. Any drop is a leak

B The change in ambient temperature, since pressure falls as the gas cools

C Whether the nitrogen has leaked through the pipe walls

D Whether the GWP has changed

15.

RECOVERY & DISPOSAL

A recovery cylinder must never be filled beyond what proportion of its capacity?

A 95%

B 50%

C 80% by weight, checked on scales

D 100%

16.

RECOVERY & DISPOSAL

When must refrigerant be recovered from a system?

A Only when the equipment is scrapped

B Only when the charge exceeds 3 kg

C Never, if the system is under 5 tonnes CO2e

D Before any work that opens the circuit, before decommissioning and before disposal of the equipment

17.

RECOVERY & DISPOSAL

What does the float switch on a recovery cylinder do?

A Stops the recovery machine when the cylinder reaches 80% full

B Indicates the refrigerant type

C Vents excess pressure

D Measures the vacuum level

18.

INSTALLATION & COMMISSIONING

Why is oxygen-free nitrogen purged through the pipework while brazing?

A To make the brazing rod flow more easily

B To stop copper oxide scale forming inside the pipe, which would contaminate the system and block filters and expansion valves

C To cool the joint faster

D To stop the flame blowing out

19.

INSTALLATION & COMMISSIONING

To what vacuum level should a system be evacuated before charging?

A Below 1 bar

B Below 5000 microns

C Below 500 microns (about 0.67 mbar), then held to check it does not rise

D Below 29 inches on the manifold gauge

20.

INSTALLATION & COMMISSIONING

What is the correct order of commissioning tasks on a new split system?

A Charge, evacuate, pressure test, run

B Evacuate, pressure test, charge, run

C Run, charge, pressure test, evacuate

D Strength and tightness test with OFN, evacuate, charge by weight, run and check superheat and subcooling

21.

INSTALLATION & COMMISSIONING

Which brazing rod can be used on copper-to-copper joints without flux?

A Silver-copper-phosphorus alloy (e.g. 2% to 15% silver)

B High-silver solder with flux is always needed

C Soft lead solder

D Aluminium rod

22.

SYSTEM OPERATION

In a refrigeration system, which component absorbs heat and makes the refrigerant cold?

A The receiver

B The evaporator, where low-pressure liquid boils and absorbs latent heat

C The condenser

D The compressor

23.

SYSTEM OPERATION

What is superheat and where is it measured?

A The liquid temperature below saturation at the condenser outlet

B The difference between room and evaporator temperature

C The temperature of the vapour above its saturation temperature, measured at the evaporator outlet or suction line

D The temperature of the compressor discharge

24.

SYSTEM OPERATION

What is latent heat?

A Heat that changes the temperature of a substance

B Heat produced by the compressor motor

C Heat lost through pipe insulation

D Heat added or removed during a change of state at constant temperature

25.

SYSTEM OPERATION

Which of these components are on the high side of the system?

A Condenser, receiver, liquid line, filter drier and sight glass

B Evaporator and suction line

C Suction accumulator

D Compressor suction valve

26.

HEALTH & SAFETY

What is the main hazard of a large refrigerant leak in a plant room or cold store?

A Electric shock

B Asphyxiation, because the vapour is heavier than air and displaces oxygen at low level

C Fire from all refrigerants

D Skin irritation

27.

HEALTH & SAFETY

Before brazing on a system that has contained refrigerant you should:

A Leave a small charge in to show up leaks

B Run the compressor to clear the pipe

C Recover the refrigerant, purge with OFN, confirm no residual pressure, remove flammables and have an extinguisher and hot work permit in place

D Open a window and start brazing

28.

HEALTH & SAFETY

Which properties make ammonia (R717) hazardous?

A It is odourless and heavier than air

B It has a very high GWP

C It is harmless but expensive

D It is toxic, mildly flammable (B2L) and attacks copper, but its strong smell gives early warning

29.

RECORDS & OPERATOR DUTIES

How long must F-gas records for equipment be kept?

A At least 5 years, by both the operator and the contractor who did the work

B 1 year

C 3 months

D Until the next leak check

30.

RECORDS & OPERATOR DUTIES

When refrigerant is added to a system on site, what must the engineer do to the equipment label?

A Write the date only

B Update it with the new total charge in kg and CO2e

C Nothing, the original label stays

D Remove it

XI Answers and explanations

1 **A.** To reduce emissions of fluorinated greenhouse gases and limit their use

The F-Gas Regulation (EU 517/2014, kept in GB law after Brexit) exists to cut emissions of fluorinated greenhouse gases such as HFCs. Ozone protection is the job of the separate ODS Regulation.

2 **B.** Leak checking only, without breaking into the refrigerant circuit

Category IV is leak checking only, and only by methods that do not break into the circuit. Category III is recovery from small systems.

3 **C.** A quota system limits the amount of HFC, measured in CO2 equivalent, that producers and importers may place on the market each year

The phase-down works through quotas on producers and importers, measured in tonnes CO2e. The quota steps down over time, which pushes the market towards lower-GWP refrigerants and makes high-GWP gas scarce and expensive.

4 **D.** That it contains F-gas, the refrigerant designation, the charge in kg and its CO2 equivalent, and the GWP

Labels must say the equipment contains fluorinated greenhouse gases, give the refrigerant designation (e.g. R410A), the quantity in kg and tonnes CO2e, and the GWP. If gas is added on site the label must be updated.

5 **A.** All refrigerant-containing parts are made tight by welding, brazing or a similar permanent connection, and the tested leak rate is below 3 g per year

The definition requires permanent joints (brazed or welded, though capped valves and service ports are allowed) and a leak rate under 3 g per year at a quarter of the maximum allowable pressure. Such systems are given relaxed leak check thresholds when labelled as hermetically sealed.

6 **B.** 1

GWP is measured relative to CO2 over 100 years, so CO2 has a GWP of 1 by definition. Everything else is compared against it.

7 **C.** 675

R32 has a GWP of 675, roughly a third of R410A, which is why it replaced R410A in small splits. It is A2L (mildly flammable).

8 **D.** R1234yf

R1234yf and R1234ze are hydrofluoro-olefins (HFOs). They have very low GWP (around 4 or less) but are A2L mildly flammable. R410A is an HFC blend, R290 is propane and R717 is ammonia.

9 **A.** The difference between the bubble point and dew point of a zeotropic blend at a given pressure

Zeotropic blends do not boil or condense at one fixed temperature. At a given pressure they start to boil at the bubble point and finish at the dew point. That spread is the glide. Use the dew point for superheat and the bubble point for subcooling.

10 **B.** Mixed refrigerant cannot be reclaimed, must be sent for destruction, and gives unpredictable pressures and performance

Mixed gas has no pressure-temperature data, can damage the system, and cannot be separated economically. It has to be destroyed at cost. Always identify the refrigerant before recovery and use a dedicated cylinder.

11 **C.** 12 months (24 months if a fixed leak detection system is fitted)

The intervals are: 5 tonnes or more every 12 months, 50 tonnes or more every 6 months, 500 tonnes or more every 3 months. A fixed leak detection system doubles the allowed interval in each band.

12 **D.** 1 month

The regulation requires a follow-up leak check within one month of a repair to verify it was effective. Ideally the repaired joint is also tested before recharging.

13 **A.** Every 12 months (12 x 2088 = 25 tonnes CO2e)

12 kg x 2088 = 25,056 kg = 25.06 tonnes CO2e. That is in the 5 to 50 tonne band, so every 12 months.

14 **B.** The change in ambient temperature, since pressure falls as the gas cools

Gas pressure changes with temperature, so a reading taken cold in the morning will be lower than one taken warm. Compare temperature-corrected readings over a long enough period before concluding.

15 **C.** 80% by weight, checked on scales

Liquid expands when warmed. A cylinder filled above 80% can become hydraulically full and rupture. Weigh the cylinder on scales during recovery and stop at 80% of its rated capacity (or rely on the float switch as a backup only).

16 **D.** Before any work that opens the circuit, before decommissioning and before disposal of the equipment

Recovery is required whenever the refrigerant would otherwise be released: opening the circuit for repair, decommissioning, and before equipment is scrapped. A certified person must do it.

17 **A.** Stops the recovery machine when the cylinder reaches 80% full

The float switch is a backup to prevent overfilling. It should not replace weighing the cylinder on scales.

18 **B.** To stop copper oxide scale forming inside the pipe, which would contaminate the system and block filters and expansion valves

Heating copper in air forms black oxide scale on the inside of the pipe. Scale breaks off and blocks driers, strainers and expansion valves. A gentle OFN purge displaces the air and keeps the pipe clean.

19 **C.** Below 500 microns (about 0.67 mbar), then held to check it does not rise

Industry practice is to pull below 500 microns and hold. If the vacuum rises quickly there is a leak; if it creeps up slowly and stalls there is still moisture boiling off. A manifold compound gauge is not accurate enough.

20 **D.** Strength and tightness test with OFN, evacuate, charge by weight, run and check superheat and subcooling

Pressure test first (leaks are found before the gas goes in), then evacuate to remove air and moisture, then weigh in the charge, then run the system and verify performance.

21 **A.** Silver-copper-phosphorus alloy (e.g. 2% to 15% silver)

Copper-phosphorus rods are self-fluxing on copper. Joining copper to brass or steel needs a high-silver alloy with a suitable flux, and the flux residue must be cleaned off afterwards.

22 **B.** The evaporator, where low-pressure liquid boils and absorbs latent heat

After the expansion device the refrigerant is a cold, low-pressure liquid. In the evaporator it boils, taking latent heat from the air or product. The condenser does the opposite, rejecting heat as the vapour turns back to liquid.

23 **C.** The temperature of the vapour above its saturation temperature, measured at the evaporator outlet or suction line

Superheat = actual suction line temperature minus the saturated evaporating temperature (found from the suction pressure on a PT chart). It proves all the liquid has boiled before the compressor. Typically 5 to 10 K.

24 **D.** Heat added or removed during a change of state at constant temperature

Latent heat is absorbed when a liquid boils and released when a vapour condenses, with no change in temperature. Sensible heat changes temperature. Refrigeration relies on latent heat because it moves far more energy per kilogram.

25 **A.** Condenser, receiver, liquid line, filter drier and sight glass

The high side runs from the compressor discharge through the condenser, receiver, liquid line, drier and sight glass to the expansion device. After the expansion device everything is low side until the compressor suction.

26 **B.** Asphyxiation, because the vapour is heavier than air and displaces oxygen at low level

Most refrigerants are heavier than air and odourless. They collect in low areas and displace oxygen. Ventilate, use an oxygen monitor or fixed detection, and never enter a suspected leak area alone.

27 **C.** Recover the refrigerant, purge with OFN, confirm no residual pressure, remove flammables and have an extinguisher and hot work permit in place

Brazing a pressurised line can blow molten metal out of the joint and decompose the refrigerant into toxic gas. Recover, make safe, purge with OFN and follow hot work procedures.

28 **D.** It is toxic, mildly flammable (B2L) and attacks copper, but its strong smell gives early warning

Ammonia is classified B2L. It is toxic at low concentrations but its pungent smell means leaks are noticed quickly. It cannot be used with copper pipe or windings, so ammonia systems use steel.

29 **A.** At least 5 years, by both the operator and the contractor who did the work

Both the operator and the undertaking (contractor) must keep records for at least five years and make them available to the regulator on request.

30 **B.** Update it with the new total charge in kg and CO2e

The label must reflect the actual quantity of F-gas in the equipment, so after adding gas the label (or an additional label) must be updated with the new charge.

XII Numbers to memorise

If you learn one page, learn this one.

3 kg / 6 kg hermetic	Charge limits for Category II and III work
3 g per year	Maximum leak rate to count as hermetically sealed
5 / 50 / 500 t CO2e	Leak check every 12 / 6 / 3 months (24 / 12 / 6 with detection)
10 t CO2e	Leak check threshold for labelled hermetic equipment
500 t CO2e	Fixed leak detection system compulsory
1 month	Re-check after a leak repair
12 months	Check interval for leak detection systems and electronic detectors
5 g per year	Minimum sensitivity of an electronic leak detector
5 years	F-gas records kept by operator and contractor
3 years	Hazardous waste consignment notes kept
80%	Maximum recovery cylinder fill, by weight
500 microns	Target vacuum before charging
1.1 × PS	Strength test pressure with OFN; tightness test at PS
5 to 10 K / 3 to 8 K	Typical superheat / subcooling
40 °C	Maximum for warming a cylinder (no flames)
25 kW	Compressor motor power above which a PSSR written scheme applies
2010 / 2015	Virgin R22 banned / recycled R22 banned for service
2020	No new stationary refrigeration ≥2500 GWP; no virgin ≥2500 GWP servicing in ≥40 t CO2e systems
2025	Single splits under 3 kg: no refrigerant ≥750 GWP
2030	Recycled and reclaimed ≥2500 GWP also banned for service
675 / 1430 / 1774 / 1810 / 2088 / 3922	GWP of R32 / R134a / R407C / R22 / R410A / R404A
1 / 0	GWP of CO2 / ODP of all HFCs and HFOs

XII Glossary

The trade's words in plain English. The same 94 terms are tappable in the practice app.

A1

Safety class: lower toxicity and no flame propagation. Examples R134a, R410A, R404A. Still an asphyxiation and frostbite hazard.

A2L

Safety class: A = lower toxicity, 2L = lower flammability with a slow burning velocity. Examples R32, R1234yf, R454B. Needs ignition sources removed and equipment rated for A2L.

A3

Safety class: lower toxicity but higher flammability. Hydrocarbons such as R290 (propane) and R600a (isobutane). Strict charge limits and spark-free equipment.

Absolute pressure (bar absolute, bar a)

Pressure measured from a perfect vacuum (zero). Absolute pressure = gauge pressure + atmospheric pressure (about 1.013 bar). Pressure-temperature charts and vacuum levels use absolute pressure, so a gauge reading of 0 bar is about 1 bar absolute.

ADR

The European agreement on carrying dangerous goods by road. Refrigerant cylinders in a vehicle need correct labelling, securing, documents and in some cases driver training.

Ambient (ambient temperature)

The temperature of the surrounding air. Head pressure rises with ambient on an air-cooled condenser, and nitrogen test pressures change with it.

AR4 (IPCC AR4)

The IPCC Fourth Assessment Report (2007). Its 100-year GWP values are the ones written into the F-Gas Regulation and used for CO2e calculations.

Asphyxiation

Suffocation from lack of oxygen. Refrigerant vapour is heavier than air and odourless, so a leak in a plant room or pit displaces the oxygen at low level without warning.

Azeotropic (azeotrope)

A blend that behaves like a single refrigerant, boiling and condensing at one temperature with no glide. Numbered in the R500 series (R507A, R502).

B2L

Safety class: higher toxicity and lower flammability. Ammonia (R717).

Brazing (braze, brazed)

Joining copper pipe with a filler rod that melts above 450 degrees C (silver or copper-phosphorus alloy), using an oxy-acetylene or similar torch. Not soft soldering.

Bubble point

For a blend at a given pressure, the temperature at which the first bubble of vapour forms when heating liquid. Use it when calculating subcooling.

CFC (CFCs, chlorofluorocarbon)

Chlorofluorocarbon. The original refrigerants such as R12 and R11, with very high ODP and GWP. Long banned.

CO2 equivalent (CO2e, tonnes CO2e)

The charge of refrigerant expressed as the amount of CO2 with the same warming effect. kg of refrigerant x GWP = kg CO2e; divide by 1000 for tonnes. Leak check and record thresholds (5, 50 and 500 tonnes) use this figure.

Company certificate (company certification, REFCOM)

A certificate held by a business (not an individual) from an approved body such as REFCOM, Quidos or Bureau Veritas, confirming it has certified staff, the right tools and procedures. Required to carry out F-gas work.

Compound gauge

The blue low-side gauge on a manifold. It reads both vacuum (below atmospheric) and pressure, hence compound. Not accurate enough to measure a deep vacuum.

Compressor

The pump of the system. Draws in low-pressure vapour from the evaporator and discharges it at high pressure and temperature so it can condense.

Condenser

The heat exchanger where high-pressure vapour from the compressor gives up its heat and condenses to liquid. This is the hot coil, outdoors on an air conditioner.

Consignment note

The document that travels with hazardous waste from producer to carrier to disposal site. Copies are kept for three years.

Copper plating

Copper dissolved by acid in a wet system deposits itself on hot steel bearing surfaces inside the compressor, causing wear and seizure. A sign of moisture contamination.

COSHH

Control of Substances Hazardous to Health Regulations. Requires employers to assess and control the risks from chemicals, including refrigerants, using information from the safety data sheet.

Decommissioning (decommission, decommissioned)

Taking equipment permanently out of service. The refrigerant must be recovered by a certified person and the recovery and disposal recorded.

Dew point

For a blend at a given pressure, the temperature at which the last drop of liquid evaporates. Use it when calculating superheat.

Direct method (direct methods, direct leak)

Leak checking the pipework itself: electronic detector, bubble solution, UV dye, or a nitrogen standing pressure test.

EN 378

The European standard for the safety and environmental requirements of refrigerating systems and heat pumps. Covers

pressure testing, charge limits for flammable refrigerants and plant room requirements.

Evacuation (evacuate, evacuating)

Pulling a vacuum on the system with a vacuum pump to remove air (non-condensables) and boil off moisture before charging. Target below 500 microns, then hold to check it does not rise.

Evaporator

The heat exchanger where low-pressure liquid refrigerant boils and absorbs heat from the air, water or product. This is the cold coil, indoors on an air conditioner.

Expansion device (expansion valve, capillary)

The component (TEV, electronic expansion valve or capillary tube) that drops the refrigerant pressure from the high side to the low side, so the liquid can boil at a low temperature in the evaporator.

F-gas (F-gases, fluorinated greenhouse gas)

Fluorinated greenhouse gas: HFCs, PFCs and SF6. They do not harm the ozone layer but have high GWPs. The F-Gas Regulation controls their use, leak checking, recovery and records.

Filter drier (drier, filter-drier)

A component in the liquid line containing desiccant to absorb moisture and acid, with a screen to trap particles. Replace it whenever the system is opened.

Flash gas

Vapour formed when liquid refrigerant suddenly drops in pressure, for example in the expansion device or in a liquid line with too little subcooling. Bubbles in the sight glass are flash gas.

Floodback (liquid floodback, slugging)

Liquid refrigerant reaching the compressor. Liquid cannot be compressed, so it damages valves and pistons and washes oil off the bearings.

Flux

A paste that cleans the joint and stops oxides forming so the filler alloy flows. Needed for copper to brass or steel; not needed with copper-phosphorus rods on copper to copper. Residue must be cleaned off.

Frostbite (cold burns)

Freezing injury to skin or eyes caused by liquid refrigerant flashing off at well below 0 degrees C. Prevented by gloves and eye protection.

Gauge pressure (bar gauge, bar g)

Pressure measured relative to the atmosphere around you, which is what a normal manifold gauge shows. 0 bar gauge means the same as atmospheric pressure. Gauge pressure = absolute pressure minus about 1.013 bar.

GWP (global warming potential)

Global Warming Potential. How much heat 1 kg of a gas traps over 100 years compared with 1 kg of CO2, which is 1. The F-Gas Regulation uses the IPCC AR4 values: R32 = 675, R134a = 1430, R407C = 1774, R410A = 2088, R404A = 3922.

Hazardous waste

Waste that is harmful to people or the environment, which includes recovered refrigerant and used refrigeration oil. Must be moved by a registered carrier to a permitted site with a consignment note.

HCFC (HCFCs, hydrochlorofluorocarbon)

Hydrochlorofluorocarbon. Contains chlorine so it depletes ozone, e.g. R22. Banned for servicing since 2015 under the ODS Regulation.

Head pressure (discharge pressure, condensing pressure)

The pressure on the high side, between the compressor discharge and the expansion device. High head pressure points to a dirty condenser, fan failure, overcharge or non-condensables.

Hermetically sealed (hermetic)

A system where every refrigerant-containing part is joined by brazing, welding or a similar permanent joint, and the tested leak rate is below 3 g per year. Service ports and capped valves are allowed. When labelled as such, the leak check threshold rises from 5 to 10 tonnes CO2e.

HFC (HFCs, hydrofluorocarbon)

Hydrofluorocarbon. Synthetic refrigerants containing hydrogen, fluorine and carbon but no chlorine, e.g. R134a, R32, R410A, R404A. Zero ODP but high GWP. These are the main F-gases.

HFO (HFOs, hydrofluoro-olefin)

Hydrofluoro-olefin. Newer synthetic refrigerants such as R1234yf and R1234ze with very low GWP (about 4 or less). Usually A2L, mildly flammable.

High side

The high-pressure part of the circuit: compressor discharge, condenser, receiver, liquid line, drier and sight glass up to the expansion device. The red gauge reads it.

Hot work permit (hot work)

A site authorisation for brazing, welding or grinding. Sets out precautions: removing flammables, extinguisher on hand, fire watch during and after the work.

HP cut-out (high pressure cut-out, high pressure (HP) cut-out)

A pressure switch that stops the compressor when discharge pressure rises too high.

Hydrogen fluoride

A highly toxic and corrosive gas produced when HFC refrigerants break down in a flame or on a very hot surface. The reason you never braze on a charged system.

Hygroscopic

Absorbs moisture from the air. POE oil is strongly hygroscopic, which is why moisture control matters so much on HFC systems.

Indirect method (indirect methods, indirect leak)

Leak checking by looking at system behaviour: pressures, temperatures, compressor current, oil and liquid levels, sight glass. A change suggests a leak which is then confirmed by a direct method.

Latent heat

Heat absorbed or released while a substance changes state (liquid to vapour or back) with no change in temperature. The evaporator absorbs latent heat as refrigerant boils; the condenser rejects it as vapour condenses.

Leak detection system (fixed leak detection, leak detection systems)

A fixed, calibrated device that monitors for refrigerant in the air or system and alerts the operator. Mandatory at 500 tonnes CO2e and above; where fitted it doubles the allowed leak check interval. Must be checked every 12 months.

Low side
The low-pressure part of the circuit: from the expansion device through the evaporator and suction line to the compressor inlet. The blue compound gauge reads it.

LP cut-out (low pressure cut-out, low pressure (LP) cut-out)
A pressure switch that stops the compressor when suction pressure falls too low, protecting it on loss of charge or a blockage.

Maximum allowable pressure (PS)
PS, the maximum pressure the system or component is designed and rated for, stated by the manufacturer. Strength tests are done at 1.1 x PS and tightness tests at PS.

Micron (microns)
A unit of very low absolute pressure used for vacuum work. 1000 microns = 1 torr = 1.33 mbar. 500 microns is about 0.67 mbar absolute. Atmospheric pressure is about 760,000 microns.

Mineral oil
Traditional compressor oil used with CFC and HCFC refrigerants such as R22. Does not mix with HFCs, so HFC systems use POE oil.

Natural refrigerant (natural refrigerants)
Refrigerants that occur in nature: hydrocarbons like R290 (propane) and R600a (isobutane), R744 (CO_2) and R717 (ammonia). Very low GWP but each has its own hazard: flammability, high pressure or toxicity.

Non-condensables (non-condensable)
Gases such as air or nitrogen that will not condense in the condenser. They collect there, raise head pressure above the saturation pressure and reduce efficiency. Removed by proper evacuation.

ODP (ozone depletion potential, ozone depleting)
Ozone Depletion Potential. How much a gas damages the ozone layer compared with R11 (CFC-11) at 1.0. Refrigerants containing chlorine or bromine (CFCs, HCFCs) have an ODP; HFCs and HFOs have an ODP of zero.

ODS (ozone depleting substance, ozone depleting substances)
Ozone Depleting Substance. Refrigerants such as R22 (HCFC) and R12 (CFC) that contain chlorine. Controlled by the ODS Regulation, separate from the F-Gas Regulation.

OFN (oxygen-free nitrogen, oxygen free nitrogen)
Oxygen-Free Nitrogen. Dry, inert gas used for pressure testing, purging while brazing and breaking a vacuum. Always used through a regulator because the cylinder is at around 200 bar.

Operator
Under the F-Gas Regulation, the person or company with actual control over the technical functioning of the equipment, normally the owner. Responsible for leak checks, records and using certified people.

Oxide scale (copper oxide scale, copper oxide)
Black flaky copper oxide that forms inside pipe when it is brazed with air in it. It breaks off and blocks driers, strainers and expansion valves. Prevented by purging with OFN.

Personnel certificate (personnel certificates)
An individual's F-gas qualification, such as City and Guilds 2079-11 Category I. Shows the person is competent to do the activities covered by that category.

Phase-down (quota)
The stepped reduction in the amount of HFC (in tonnes CO_2e) that producers and importers may place on the market each year, enforced through quotas. It makes high-GWP gas scarce and expensive.

POE oil (POE, polyol ester)
Polyol ester oil, used with HFC refrigerants. It is hygroscopic (absorbs moisture from the air), so keep containers sealed and systems open for as short a time as possible.

PSSR (Pressure Systems Safety Regulations)
Pressure Systems Safety Regulations 2000. Refrigeration systems with compressor motor power above 25 kW need a written scheme of examination and periodic inspection by a competent person.

PT chart (pressure-temperature chart, pressure temperature)
A pressure-temperature chart or slide rule showing the saturation temperature of each refrigerant at each pressure. Used to work out superheat and subcooling from gauge readings.

Pump-down (pump down)
Running the compressor with the liquid line valve closed to collect the charge in the receiver or condenser, so the low side can be opened for service without recovering the whole charge.

Push-pull (push-pull method)
A fast recovery method for large liquid charges. The recovery machine pulls vapour from the top of the cylinder and pushes it into the system, forcing liquid out of the system into the cylinder.

Receiver (liquid receiver)
A vessel after the condenser that stores liquid refrigerant and allows for changes in charge demand. Also used to hold the charge during pump-down.

Reclamation (reclaimed, reclaim)
Reprocessing recovered refrigerant at a specialist facility to meet a virgin product standard (e.g. AHRI 700) so it can be sold and used anywhere.

Recovery (recover, recovered)
Removing refrigerant from a system into a recovery cylinder using a recovery machine, so it is not released to atmosphere. Required before opening the circuit, decommissioning or scrapping.

Recovery cylinder (recovery cylinders)
A refillable pressure cylinder, grey with a yellow shoulder in the UK, dedicated to one refrigerant and in test date. Never filled beyond 80% by weight.

Recycling (recycled, recycle)
Basic cleaning of recovered refrigerant (oil separation, filtering, moisture removal) so it can be reused, normally in equipment belonging to the same owner.

Reversing valve (four-way valve)
The four-way valve in a heat pump that redirects the compressor discharge so the indoor coil becomes the condenser (heating) or the evaporator (cooling).

Safety data sheet (SDS)
The supplier's document for a substance giving its hazards, handling, storage, PPE, first aid and disposal information.

Saturation temperature (saturated evaporating temperature, saturated condensing temperature)

The temperature at which a refrigerant boils or condenses at a given pressure. Fix the pressure and you fix the temperature. A PT chart lists this relationship for each refrigerant.

Schrader valve (Schrader, Schrader core)

A spring-loaded service valve like a tyre valve, used at gauge ports. The core is for service access only; the screw-on cap with its seal is the primary leak seal.

Sensible heat

Heat that changes the temperature of something without changing its state, such as warming air or cooling liquid refrigerant below saturation.

Service ban (2020 service ban)

Since 1 January 2020, virgin HFCs with a GWP of 2500 or more may not be used to service equipment with a charge of 40 tonnes CO2e or more. Recycled or reclaimed gas is allowed until 2030.

Sight glass

A small window in the liquid line. Clear glass means a full column of liquid; bubbles mean flash gas, usually from low charge or a restriction. Many include a moisture indicator that changes colour when wet.

Strength test (strength (pressure) test, pressure test)

A test with OFN at 1.1 times the maximum allowable pressure to prove the pipework and joints will not burst. Done before the tightness test and before any refrigerant is added.

Subcooling (subcool, subcooled)

How many degrees a liquid is below its saturation temperature. Measured at the condenser outlet: saturated condensing temperature from the discharge pressure minus the actual liquid line temperature. Proves solid liquid is reaching the expansion device. Typically 3 to 8 K.

Suction accumulator (accumulator)

A vessel in the suction line that catches any liquid refrigerant before it reaches the compressor, letting it boil off safely.

Suction pressure (evaporating pressure)

The pressure on the low side, in the evaporator and suction line. Low suction pressure points to low charge, a restriction or a blocked evaporator.

Superheat (superheated)

How many degrees a vapour is above its saturation temperature. Measured at the evaporator outlet: actual suction line temperature minus the saturated evaporating temperature from the suction pressure. Proves all liquid has boiled off before the compressor. Typically 5 to 10 K.

Temperature glide (glide)

The temperature range over which a zeotropic blend boils or condenses at a fixed pressure: the difference between its bubble point and dew point. R407C has a glide of about 5 to 7 K; R410A is under 0.2 K.

TEV (thermostatic expansion valve, TXV)

Thermostatic Expansion Valve. Meters liquid into the evaporator and controls superheat using a sensing bulb clamped to the evaporator outlet.

Tightness test (standing pressure test, standing pressure)

A leak test with OFN, usually at the maximum allowable pressure, held long enough (often 24 hours) to show no pressure drop once temperature changes are allowed for.

Triple evacuation

Pull a vacuum, break it with OFN, pull again, break again, pull a third time. The dry nitrogen absorbs residual moisture, which is removed on the next pull. Used on wet or contaminated systems.

Undertaking

The regulation's word for a business or company. An undertaking carrying out F-gas work needs a company certificate, as well as certified engineers.

Virgin refrigerant (virgin)

New, unused refrigerant straight from the manufacturer, as opposed to recovered, recycled or reclaimed gas.

Written scheme of examination

A document drawn up by a competent person listing which parts of a pressure system must be inspected, how and how often, as required by PSSR.

Zeotropic (zeotrope)

A blend whose components boil at different temperatures, so it has a temperature glide. Numbered in the R400 series (R407C, R410A, R404A). Must be charged as liquid so the mix stays correct.

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